



YEAR 7 MATHEMATICS · CHAPTER 10

# Comparing data sets

Lesson 8 of 8

**In this lesson you will learn to:**

- compare two data sets using measures of centre and spread
- draw and read a back-to-back stem-and-leaf plot
- tell the difference between primary and secondary data

Name

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Class

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Teacher

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## Before you start

You will need

Pencil

Ruler

Calculator

**Syllabus:** MA4-DAT-C-02: analyses simple datasets using measures of centre, range and shape of the data

### WE DO

**Grey boxes.** Work with your teacher. Copy what they write on the board.

### YOU DO

**Blue boxes.** Work on your own. Show your working in the box.

- 1 Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2 At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

## Lesson map

| Page | What is on it                                                               | Done                     |
|------|-----------------------------------------------------------------------------|--------------------------|
| 3    | Learning intentions, key terms, notes and Level 1 teacher examples          | <input type="checkbox"/> |
| 4    | Level 1: <b>YOU DO</b> practice questions                                   | <input type="checkbox"/> |
| 5    | Level 2: <b>WE DO</b> teacher examples                                      | <input type="checkbox"/> |
| 6    | Level 2: <b>YOU DO</b> practice questions                                   | <input type="checkbox"/> |
| 7    | Level 3: <b>WE DO</b> teacher examples and <b>YOU DO</b> practice questions | <input type="checkbox"/> |
| 8    | Level 3: <b>YOU DO</b> practice questions                                   | <input type="checkbox"/> |
| 9    | Extension questions and the tear-off exit ticket                            | <input type="checkbox"/> |
| 10   | Lesson summary: a worked example for every level                            | <input type="checkbox"/> |

### YOU DO

**On your own: Warm-up. These skills will help you in this lesson. Show your working.**

- 1 Median of 11, 14, 12, 18, 15

- 3 Mean of 5 and 15

- 2 Range: lowest 145, highest 175

- 4 A bigger range means the data is more ...



### Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

# LESSON 10.08

## Comparing data sets

Date \_\_\_\_\_

### Learning intentions

We are learning to:

- compare two data sets using measures of centre and spread
- draw and read a back-to-back stem-and-leaf plot
- tell the difference between primary and secondary data

### Success criteria

I can:

- ☐ compare the **mean** or **median** to decide which group did better
- ☐ compare the **range** to decide which group is more consistent
- ☐ draw a **back-to-back stem-and-leaf plot** and write a conclusion

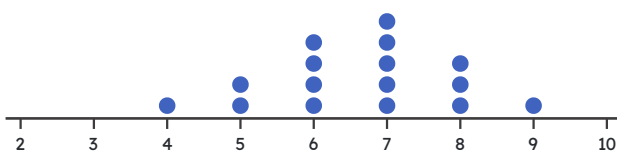
**Key terms** Write the definition as your teacher explains it.

**Back-to-back stem-and-leaf plot**
**Primary data**
**Secondary data**

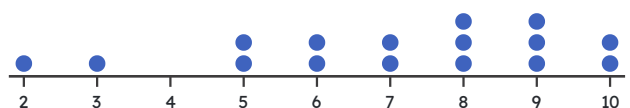
Notes

### Level 1 · Getting started

Class A quiz scores



Class B quiz scores



### WE DO

With your teacher: copy what they write on the board.

### Example 1

Find the mode and the range of each class.

### Example 2

Find the median of each class. Which class did better? Which was more consistent?

**YOU DO** On your own: Set A. Ages of people at two gyms.

| Gym A   | Stem | Gym B     |
|---------|------|-----------|
| 8       | 1    |           |
| 7 5 5 2 | 2    | 1 4 9     |
| 8 4 3 1 | 3    | 3 5 5 6 9 |
| 5 2     | 4    | 1 4 6 8   |
| 1       | 5    | 2 7       |
|         | 6    | 3         |

Key: 3 | 5 = 35 years

- 1 How many people go to Gym A?
- 2 How many people go to Gym B?
- 3 Who is the youngest at Gym A?
- 4 Who is the oldest at Gym B?
- 5 What is the mode for Gym A?
- 6 What is the mode for Gym B?
- 7 What is the range for Gym A?
- 8 What is the range for Gym B?
- 9 How many people are in their 30s altogether?
- 10 Which gym has older people in general?

**YOU DO** On your own: Set B. Is each one primary data (you collect it yourself) or secondary data (someone else collected it)? Write P or S.

- |                                                                                   |                                                                         |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 You measure the heights of your class. <input type="text"/>                     | 6 You copy a population table from a textbook. <input type="text"/>     |
| 2 You download rainfall data from the Bureau of Meteorology. <input type="text"/> | 7 You time your friends running 100 m. <input type="text"/>             |
| 3 You hand out a survey at lunch. <input type="text"/>                            | 8 You use sports results from a news website. <input type="text"/>      |
| 4 You use Census results from the ABS. <input type="text"/>                       | 9 You toss a coin 50 times and record the results. <input type="text"/> |
| 5 You count cars driving past the school gate. <input type="text"/>               | 10 You use data from a science journal. <input type="text"/>            |

 **Level 2 · Building up**

**WE DO** With your teacher: copy what they write on the board.

**Spelling scores.** Class P: 34, 41, 45, 28, 39, 47, 52, 36    Class Q: 25, 31, 38, 29, 42, 33, 27, 35

**Example 3** Draw a back-to-back stem-and-leaf plot of the scores.

| Class P | Stem | Class Q |
|---------|------|---------|
|         | 2    |         |
|         | 3    |         |
|         | 4    |         |
|         | 5    |         |

Key: \_\_\_\_ | \_\_\_\_ = \_\_\_\_\_

**Example 4** Find the median and range of each class. Compare the two classes.

**YOU DO**

On your own: Set C. Heights (cm) of two Under-14 basketball squads.

**Girls:** 163, 157, 170, 158, 152, 166, 160, 148, 161, 161, 155**Boys:** 156, 150, 145, 158, 158, 158, 171, 153, 167, 149, 162, 175

- 1 Draw a back-to-back stem-and-leaf plot. Put the girls on the left.

| Girls | Stem | Boys |
|-------|------|------|
|       | 14   |      |
|       | 15   |      |
|       | 16   |      |
|       | 17   |      |

Key: \_\_\_\_ | \_\_\_\_ = \_\_\_\_\_

- 2 Who is the tallest player? Which squad are they in?

- 5 Find the range of each squad.

- 6 Find the mean of each squad (2 decimal places).

- 7 Which squad is taller on average? Use a statistic.

- 8 Which squad has heights that are more alike? Use a statistic.

- 3 Find the median height of each squad.

- 4 Find the mode of each squad.

- 9 How many players in each squad are 160 cm or taller?

- 10 Write two sentences comparing the squads.

- 11 The coach measured the players herself. Is this primary or secondary data?

- 12 A coach wants the taller squad for a tournament. Which squad would you pick? Use two statistics.



## Level 3 · Challenge

**WE DO** With your teacher: copy what they write on the board.

**Goals in 6 netball games.** Amy: 12, 14, 13, 15, 14, 12    Beth: 5, 22, 8, 20, 18, 11

**Example 5** Find the mean and the range for each player. Who is more consistent?

**Example 6** Which player would you pick for the grand final? Use the statistics to justify your choice.

**YOU DO** On your own: Set D. Assignment marks (out of 100) for two classes.

**7R:** 52, 78, 72, 72, 86, 67, 61, 68, 81, 75, 63, 70, 64, 45, 58

**7T:** 66, 79, 92, 60, 74, 55, 23, 49, 88, 95, 83, 38, 90, 71, 97

- ① Draw a back-to-back stem-and-leaf plot. Put 7R on the left.

| 7R | Stem | 7T |
|----|------|----|
|    | 2    |    |
|    | 3    |    |
|    | 4    |    |
|    | 5    |    |
|    | 6    |    |
|    | 7    |    |
|    | 8    |    |
|    | 9    |    |

Key: \_\_\_\_ | \_\_\_\_ = \_\_\_\_\_

- ② What was the highest mark? Which class scored it?

- ③ Find the mean, median, mode and range of 7R.

- ④ Find the mean, median, mode and range of 7T.

**YOU DO****On your own: Set D continued. Use your plot and statistics from the previous page.**

- 5 Which class did better overall? Use the mean or median to explain.

- 8 Remove that mark. Find the new mean and range for 7T.

- 6 Which class was more consistent? Use the range to explain.

- 9 The teacher got the marks from her own mark book. Is this primary or secondary data? Explain.

- 7 Which mark in 7T could be called an outlier? Why?

- 10 Write a conclusion comparing 7R and 7T using at least two statistics.



## Extension · Stretch yourself

Finished Level 3? Stretch yourself here.



## Worked example: combining groups

Class A has 10 students with a mean of 70. Class B has 15 students with a mean of 80. What is the mean of all 25 students?

1. Total for A:  $10 \times 70 = 700$
2. Total for B:  $15 \times 80 = 1200$
3. Total for everyone:  $700 + 1200 = 1900$
4. Mean:  $1900 \div 25 = 76$

**Answer:** 76 (not 75, because Class B is bigger)

## YOU DO

On your own: read the worked example, then try these.

- E1** A group of 8 has a mean of 12 and a group of 12 has a mean of 17. Find the mean of all 20.

- E2** Why is the combined mean not always halfway between the two means?

- E3** Make up two sets of 5 values with the same mean but different ranges.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

## YOU DO

## Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

| Team A | Stem | Team B    |
|--------|------|-----------|
| 5 2    | 1    | 8         |
| 4 4 1  | 2    | 2 5 7 8 9 |
| 0      | 3    |           |

Key: 2 | 4 = 24

## Level 1

What is the highest score? Which team scored it?

## Level 2

Find the median of each team.

## Level 3

Which team was more consistent? Explain using the range.

## Lesson summary

Missed the lesson? Start here.

## Comparing data sets

- Centre** (mean or median): which group is higher?
- Spread** (range): smaller range = more consistent.
- Look for **outliers** that change the mean.
- Primary** data you collect; **secondary** data someone else collected.

## Level 1

In this back-to-back plot, what does the left leaf 3 on stem 2 stand for?

- Left leaves belong to the left group (Team A)
- Stem 2 and leaf 3 make 23

| Team A | Stem | Team B |
|--------|------|--------|
| 7 3    | 2    | 1 5    |
| 4      | 3    | 0 2 6  |

Key: 2 | 3 = 23

**Answer:** a score of 23 for Team A

## Level 2

Medians: Class X 64, Class Y 71. Ranges: X 20, Y 45. Compare the classes.

- Y has the higher median, so Y did better on average
- X has the smaller range, so X is more consistent

**Answer:** Y scored higher; X was more consistent

## Level 3

Amy: mean 13.3, range 3. Beth: mean 14, range 17. Who is more reliable?

- The means are close
- Amy's range is much smaller, so her scores are more alike

**Answer:** Amy is more reliable

## Extension

4 values have a mean of 5 and 6 values have a mean of 10. What is the combined mean?

- Totals:  $4 \times 5 = 20$  and  $6 \times 10 = 60$
- Combined:  $80 \div 10 = 8$

**Answer:** 8 (not 7.5, because the second group is bigger)

This space is left blank on purpose. It is the back of your exit ticket.